

Repo 5 — PD Model Feature Dictionary



Enterprise Probability of Default (PD) Feature Documentation

This document describes the engineered and selected features used for the Probability of Default (PD) model in Repo 5.

The PD model is designed to predict the probability that a borrower defaults after loan origination using only information available at origination time (no future leakage).

PD Feature Dictionary

Feature	Formula / Engineering Logic	Business Meaning	Why Important for PD
loan_amnt	Raw feature	Requested loan amount	Larger loans may increase repayment burden and exposure risk
funded_amnt	Raw feature	Actual funded loan amount	Represents true bank exposure
funded_amnt_inv	Raw feature	Investor-funded portion	Marketplace participation and funding structure
int_rate	Raw feature	Interest rate assigned to borrower	Reflects pricing and perceived borrower risk
installment	Raw feature	Monthly repayment amount	Measures monthly repayment burden
term_months	Extracted from <code>term</code>	Loan maturity duration	Longer-term loans carry more uncertainty and risk
annual_inc	Raw feature	Annual borrower income	Core repayment capacity indicator

Feature	Formula / Engineering Logic	Business Meaning	Why Important for PD
emp_length_years	Extracted numeric years from emp_length	Employment stability	Stable employment generally reduces default risk
verification_status	Categorical feature	Income verification status	Verified income is more reliable
home_ownership	Categorical feature	Housing ownership structure	Mortgage ownership often correlates with financial stability
emp_title_clean	Cleaned occupation text	Standardized occupation representation	Captures occupational repayment behavior
job_cluster	Engineered occupation clusters	Economic sector segmentation	Captures occupational risk patterns
fico_range_low	Raw feature	Lower FICO score boundary	Indicates borrower credit quality
fico_range_high	Raw feature	Upper FICO score boundary	Measures borrower creditworthiness
delinq_2yrs	Raw feature	Delinquencies in last 2 years	Historical repayment problems
pub_rec	Raw feature	Public derogatory records	Severe financial distress indicator
pub_rec_bankruptcies	Raw feature	Bankruptcy history	Strong distress signal
inq_last_6mths	Raw feature	Credit inquiries in last 6 months	Measures recent credit demand / credit hunger
acc_now_delinq	Raw feature	Current delinquent accounts	Active repayment stress indicator

Feature	Formula / Engineering Logic	Business Meaning	Why Important for PD
revol_bal	Raw feature	Revolving debt balance	Existing debt burden
revol_util	Raw feature	Revolving utilization percentage	Measures revolving credit stress
bc_util	Raw feature	Bankcard utilization percentage	Measures card utilization pressure
all_util	Raw feature	Combined utilization metric	Overall debt pressure indicator
il_util	Raw feature	Installment utilization ratio	Installment borrowing pressure
revol_util_ratio	<code>revol_util / 100</code>	Normalized revolving utilization	ML-friendly utilization ratio
bc_util_ratio	<code>bc_util / 100</code>	Normalized bankcard utilization	Captures card utilization intensity
all_util_ratio	<code>all_util / 100</code>	Normalized aggregate utilization	Overall leverage intensity
installment_income_ratio	<code>installment / annual_inc</code>	Monthly debt burden relative to income	Measures affordability pressure
loan_income_ratio	<code>loan_amnt / annual_inc</code>	Loan leverage relative to income	Captures borrower leverage intensity
credit_history_months	<code>(issue_d - earliest_cr_line)</code> converted to months	Length of borrower credit history	Longer history generally indicates stability
total_acc	Raw feature	Total lifetime credit accounts	Measures overall credit depth
open_acc	Raw feature	Currently open credit accounts	Active borrowing obligations

Feature	Formula / Engineering Logic	Business Meaning	Why Important for PD
mort_acc	Raw feature	Mortgage account count	Proxy for financial maturity
num_actv_rev_tl	Raw feature	Active revolving trades	Current revolving leverage exposure
num_actv_bc_tl	Raw feature	Active bankcard trades	Current card utilization exposure
accounts_per_year	$\frac{\text{total_acc}}{(\text{credit_history_years} + 1)}$	Credit acquisition velocity	Captures aggressive credit expansion behavior
mo_sin_old_rev_tl_op	Raw feature	Months since oldest revolving trade	Measures borrower maturity
mo_sin_rcnt_rev_tl_op	Raw feature	Months since recent revolving trade	Captures recent borrowing activity
mths_since_recent_inq	Raw feature	Months since latest credit inquiry	Measures recent credit demand
mths_since_last_delinq	Raw feature	Months since last delinquency	Measures repayment recovery stability
mths_since_last_record	Raw feature	Months since last public record	Measures severity persistence
addr_state	Categorical feature	Borrower state	Captures regional economic conditions
zip_region	Extracted ZIP prefix	Geographic segmentation	Local economic and behavioral risk patterns

Feature	Formula / Engineering Logic	Business Meaning	Why Important for PD
issue_year	Extracted from <code>issue_d</code>	Economic regime period	Captures macroeconomic cycle
issue_month	Extracted from <code>issue_d</code>	Loan issuance month	Captures seasonality effects
issue_quarter	Extracted from <code>issue_d</code>	Loan issuance quarter	Captures macro-cycle timing
application_type	Categorical feature	Individual vs joint application	Joint borrowers may behave differently
annual_inc_joint	Raw feature	Combined household income	Household repayment capacity
dti_joint	Raw feature	Joint debt-to-income ratio	Household leverage burden
joint_app_flag	<code>annual_inc_joint not null</code>	Joint application indicator	Structural borrower segmentation
all_util_missing	<code>all_util is null</code>	Missing utilization indicator	Missingness itself may contain behavioral information
il_util_missing	<code>il_util is null</code>	Missing installment utilization indicator	Structural or behavioral missingness
mths_since_last_delinq_missing	<code>mths_since_last_delinq is null</code>	Missing delinquency history indicator	May indicate clean borrower history
dti	Raw feature	Debt-to-income ratio	Core leverage and affordability measure
total_rev_hi_lim	Raw feature	Total revolving credit limit	Measures borrower credit capacity
tot_cur_bal	Raw feature	Total current balance across accounts	Aggregate debt burden

Feature	Formula / Engineering Logic	Business Meaning	Why Important for PD
avg_cur_bal	Raw feature	Average balance across accounts	Average account leverage
tot_hi_cred_lim	Raw feature	Total high credit limit	Borrower overall credit capacity
total_bal_ex_mort	Raw feature	Total balance excluding mortgage	Non-housing debt burden
total_bc_limit	Raw feature	Total bankcard limit	Available card capacity
total_il_high_credit_limit	Raw feature	Total installment high credit limit	Installment borrowing capacity
num_rev_accts	Raw feature	Number of revolving accounts	Revolving debt structure
num_bc_tl	Raw feature	Number of bankcard trades	Credit card exposure complexity
num_il_tl	Raw feature	Number of installment trades	Installment borrowing exposure
pct_tl_nvr_dlq	Raw feature	Percentage of trades never delinquent	Historical repayment quality
percent_bc_gt_75	Raw feature	Percentage of bankcards above 75% utilization	High utilization stress indicator
tax_liens	Raw feature	Tax lien count	Severe financial/ legal distress indicator
tot_coll_amt	Raw feature	Total collections amount	Historical collections activity
collections_12_mths_ex_med	Raw feature	Collections in last 12 months excluding medical	Recent collection activity

Feature	Formula / Engineering Logic	Business Meaning	Why Important for PD
inq_fi	Raw feature	Personal finance inquiries	Borrower credit demand intensity
inq_last_12m	Raw feature	Credit inquiries in last 12 months	Medium-term borrowing demand
open_acc_6m	Raw feature	Newly opened accounts in last 6 months	Recent credit expansion
open_il_12m	Raw feature	Installment accounts opened in last 12 months	Installment credit growth
open_rv_12m	Raw feature	Revolving accounts opened in last 12 months	Revolving credit growth
open_rv_24m	Raw feature	Revolving accounts opened in last 24 months	Longer-term revolving expansion
acc_open_past_24mths	Raw feature	Accounts opened in past 24 months	Overall credit growth velocity

Important Modeling Notes ★★★★★

Leakage Features Removed

The following features were intentionally removed from the PD model because they contain future repayment outcomes or post-default information:

- grade
- sub_grade
- loan_status
- last_pymnt_d
- next_pymnt_d
- recoveries
- collection_recovery_fee
- settlement_* features
- hardship_* features

- total_rec_prncp
- total_rec_int
- out_prncp
- out_prncp_inv
- last_fico_range_low
- last_fico_range_high
- total_pymnt
- total_pymnt_inv
- last_pymnt_amnt

These would artificially inflate model performance and violate origination-time prediction assumptions.

Enterprise Modeling Philosophy ★★★★★

The final PD feature set intentionally combines:

- credit quality,
- utilization stress,
- leverage,
- repayment burden,
- occupational segmentation,
- geographic segmentation,
- macro timing,
- missingness behavior,
- temporal maturity,
- application structure.

This creates a significantly richer enterprise-style PD framework compared to simple Kaggle-style default prediction pipelines.